

Customer Behavioral Segmentation System

Phase 2 Report — Preprocessing, K-Means Clustering & EDA

B.Tech Artificial Intelligence & Machine Learning

Field	Details
Project Title	Customer Behavioral Segmentation System
Phase	Phase 2 — EDA, Preprocessing & K-Means Clustering
Primary Algorithm	K-Means Clustering (k=4, k-means++ init)
Secondary Clustering	K-Means on Age + Gender (k=3)
Features Used	Annual_Spending, Purchase_Frequency, Age, Gender_Encoded
Evaluation Metrics	WCSS (Elbow Method), Silhouette Score
Key Output	4 Customer Personas with Business Interpretations

Table of Contents

1. Problem Description
2. Architecture
3. Models Used
4. Evaluation Metrics
5. Preliminary Results and Visualizations
6. References

1. Problem Description

The Customer Behavioral Segmentation System addresses one of the most persistent and costly problems in retail marketing — the inability to distinguish between fundamentally different types of customers and tailor communication accordingly. A business that sends the same promotional content to all 1000 customers in its database is essentially flying blind. A customer who purchases 25 times a year and spends ₹18,000 annually has an entirely different relationship with the brand than someone who shops twice a year and spends under ₹500. Yet without a systematic segmentation framework, both receive the same message, the same discount, and the same level of attention — which means the high-value customer is underserved, and the discount-seeker is being offered too much.

Phase 2 of this project focuses on solving this problem through unsupervised machine learning. The approach begins with careful preprocessing of the raw retail dataset (`retail_finding_dataa.csv`), which contains 1000 customer records across 9 columns including Age, Gender, Region, Annual_Spending, Purchase_Frequency, Discount_Usage_Pct, Returns_Count, and Preferred_Category. EDA from Phase 1 confirmed that Annual_Spending is right-skewed (skewness 1.41) and strongly correlated with Purchase_Frequency ($r = 0.89$), making this pair an ideal candidate for primary behavioral clustering.

The central modeling task in Phase 2 is to apply K-Means clustering on the Annual_Spending and Purchase_Frequency features, after StandardScaler normalization, to discover natural groupings among customers. The Elbow Method is used to objectively determine that four clusters ($k=4$) best represent the data. These four clusters are then analyzed and assigned meaningful business personas: Budget Buyers, Occasional Shoppers, Regular Spenders, and Premium High-Value Customers. A secondary clustering on Age and Gender ($k=3$) further enriches the analysis by connecting behavioral segments to demographic groups.

The business value of this phase is concrete and immediate. Once persona groups are established and their behavioral characteristics quantified (average spending per group, purchase frequency, discount usage, return rates), the marketing team has a scientific basis for targeted campaigns. Budget Buyers need flash sale promotions. Premium High-Value Customers need VIP programs and early product access. Each group gets communication that is relevant to how they actually behave — and that is exactly what this phase delivers.

2. Architecture

The architecture of Phase 2 follows a linear preprocessing-to-clustering-to-interpretation pipeline. The flow begins with the validated dataset from Phase 1 and ends with saved persona-labeled customer records and visual analysis ready for business use.

PHASE 2 SYSTEM ARCHITECTURE

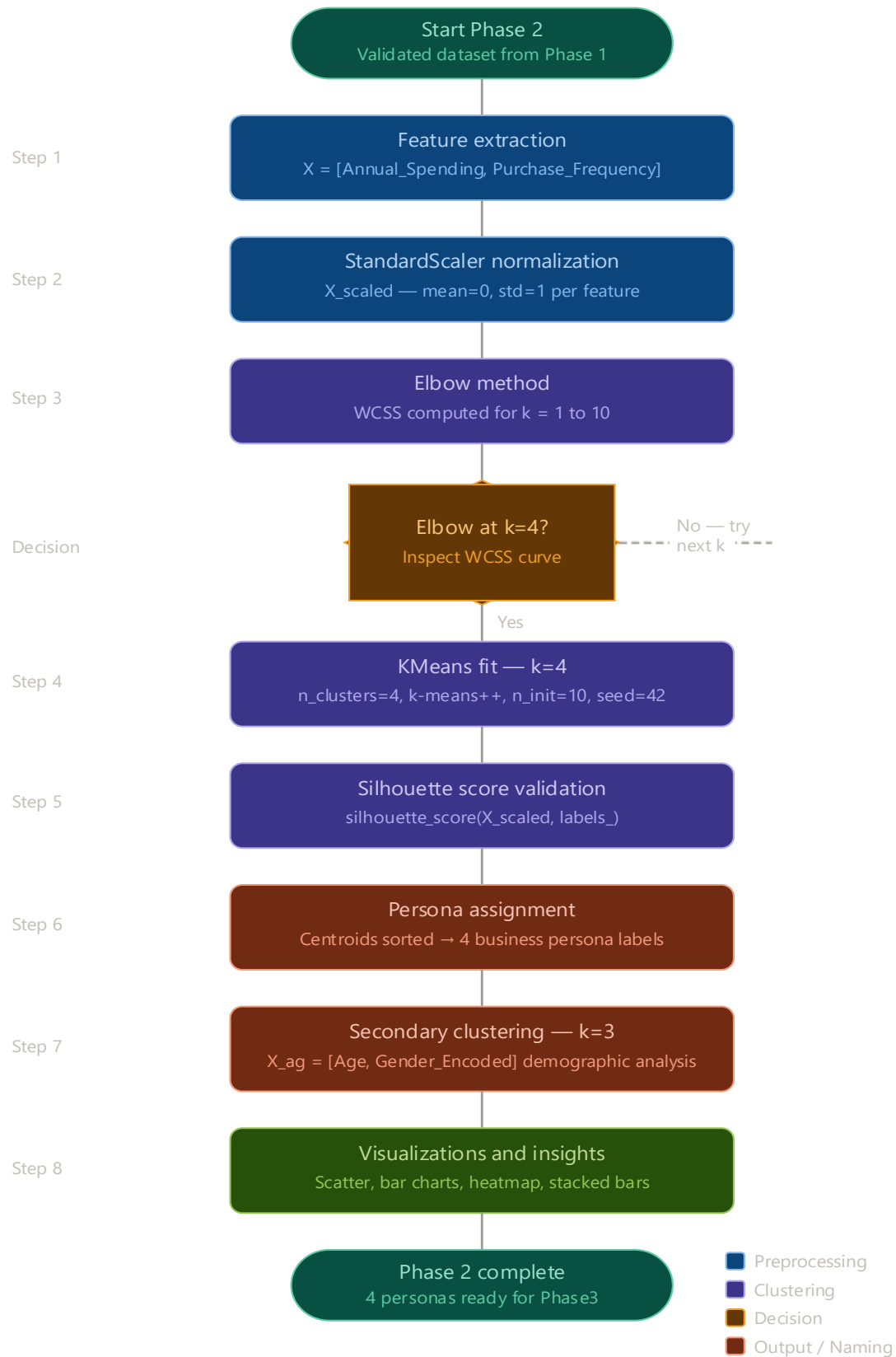


Fig. 1 — Phase 2 Architecture Flow Diagram

Each stage of this pipeline has a deliberate purpose. The feature extraction step selects only Annual_Spending and Purchase_Frequency for primary clustering, as confirmed by their strong correlation ($r=0.89$) and their direct relevance to purchasing behavior. StandardScaler is applied because K-Means uses Euclidean distance — without scaling, Annual_Spending's larger numeric range would dominate all distance calculations, effectively making Purchase_Frequency invisible to the algorithm. The Elbow Method removes subjectivity from the cluster count decision, giving a mathematically-grounded reason for choosing $k=4$. The Silhouette Score then validates that the four clusters found are genuinely distinct rather than arbitrary partitions.

3. Models Used

3.1 K-Means Clustering (Primary Model)

K-Means is a centroid-based clustering algorithm that partitions n data points into k non-overlapping groups by minimizing the total Within-Cluster Sum of Squares (WCSS). In this project, K-Means serves as the primary segmentation model because it produces clean, well-separated, and easily interpretable clusters — exactly what is needed to define business personas.

Internally, K-Means operates through an Expectation-Maximization (EM) loop. The algorithm is initialized with $k=4$ random centroids placed using the `k-means++` strategy (which spaces initial centroids far apart to avoid poor random starts). Every data point is then assigned to its nearest centroid using Euclidean distance in the scaled feature space. Centroids are then recomputed as the mean of all points assigned to them. This assignment-update loop repeats until centroid positions stop changing — convergence. The final centroids are then inverse-transformed (using `scaler.inverse_transform()`) to recover original-scale coordinates, which allows meaningful interpretation: 'Cluster 3 has an average annual spending of ₹16,500 and purchases 24 times per year.'

The specific sklearn parameters used are: `n_clusters=4`, `init='k-means++'`, `n_init=10`, `random_state=42`. Setting `n_init=10` means the algorithm runs 10 independent times with different initializations and keeps the result with lowest WCSS — this guards against convergence to a local minimum. `random_state=42` ensures the result is reproducible.

Parameter	Value	Reason
n_clusters	4	Determined by Elbow Method (WCSS elbow at $k=4$)
init	k-means++	Smarter initialization reduces poor random starts

Parameter	Value	Reason
n_init	10	Runs 10 times, picks best — reduces local minima risk
random_state	42	Reproducibility across runs

3.2 Persona Assignment Logic

After KMeans assigns numeric cluster labels (0, 1, 2, 3), the code maps these to human-readable business personas using the `persona_map` dictionary. Cluster centers are inverse-transformed to original scale and sorted by `Annual_Spending` to determine which cluster represents which tier. The resulting `persona_map` is: {0: 'Budget Buyers', 1: 'Occasional Shoppers', 2: 'Regular Spenders', 3: 'Premium High-Value Customers'}. This label assignment is what transforms a mathematical output into a business tool.

3.3 K-Means on Age + Gender (Secondary Clustering)

A secondary K-Means analysis is performed on the demographic features `Age` and `Gender_Encoded`. `Gender` is first encoded numerically using `LabelEncoder` (`le.fit_transform(df['Gender'])`), producing integer codes for Female, Male, and Non-Binary. The combined feature matrix `X_ag = df[['Age', 'Gender_Encoded']]` is then scaled and clustered with `k=3` (from a separate Elbow Method on these features). This secondary clustering surfaces demographic patterns — for example, whether younger female customers cluster separately from older male customers — and enables the cross-analysis heatmap (`Persona vs Age_Gender_Cluster`) that reveals hyper-targeted campaign opportunities.

4. Evaluation Metrics

4.1 WCSS — Within-Cluster Sum of Squares (Elbow Method)

WCSS, also referred to as inertia in scikit-learn, is the total sum of squared Euclidean distances from each data point to its assigned cluster centroid. Formally: $WCSS = \sum \text{over all clusters of } (\sum \text{of squared distances from each point in the cluster to the cluster centroid})$. A lower WCSS means points are tightly packed around their centroids — indicating compact, well-defined clusters. However, WCSS always decreases as `k` increases (in the extreme, `k=n` gives `WCSS=0`). The Elbow Method addresses this by plotting WCSS against `k` and looking for the inflection point where the rate of decrease sharply slows — the 'elbow' — indicating that beyond this `k`, additional clusters do not meaningfully improve compactness. In this project, the elbow is clearly visible at `k=4`.

4.2 Silhouette Score

The Silhouette Score provides a more complete validation of cluster quality than WCSS alone, because it measures both cohesion (how close points are within their own cluster) and separation (how far they are from the nearest other cluster). For each point i , the silhouette value is: $s(i) = (b(i) - a(i)) / \max(a(i), b(i))$, where $a(i)$ is the mean distance to all other points in the same cluster, and $b(i)$ is the mean distance to all points in the nearest different cluster. The overall Silhouette Score is the mean across all points. Scores range from -1 to +1: above 0.5 indicates good separation, above 0.7 is excellent. The code evaluates this using `sklearn.metrics.silhouette_score(X_scaled, kmeans_final.labels_)` and prints an interpretation message: 'Good clustering. Personas are well separated' if the score exceeds 0.5.

Metric	Formula / Range	Interpretation in This Project
WCSS / Inertia	Sum of squared distances to centroids. Lower is better.	Elbow at k=4 used to finalize cluster count
Silhouette Score	$(b-a)/\max(a,b)$. Range: -1 to +1	>0.5 = well-separated personas; computed on final k=4 model
Cluster Profile Table	Mean feature values per cluster	Verifies personas are behaviorally distinct, not random

5. Preliminary Results and Visualizations

5.1 EDA Findings

Before clustering, Phase 2 begins with an extended EDA that builds on Phase 1 findings. The key insights extracted are:

- Annual_Spending is right-skewed (skewness 1.41), meaning most customers are budget-to-moderate spenders, but a long tail of high spenders pulls the mean upward. This skew is what drives the need for scaling — raw K-Means on such a distribution would create unequal-sized clusters dominated by the central mass.
- Purchase_Frequency (skewness 1.56) shows a similar pattern — most customers shop a few times a year, but power buyers shop very frequently. This right tail is where premium customers live.
- The correlation between Annual_Spending and Purchase_Frequency ($r=0.89$) confirms these two features move together — more frequent buyers are also higher spenders. This strong correlation validates using them as the primary clustering pair: they jointly capture purchasing power better than either alone.
- Discount_Usage_Pct is negatively correlated with Annual_Spending (-0.38), confirming that price-sensitive, discount-dependent customers tend to spend less overall — an insight directly reflected in the Budget Buyers persona.
- Age has weak correlations with all behavioral features ($|r| < 0.15$), confirming that age alone does not predict spending behavior — which is why a separate demographic clustering analysis on Age + Gender is warranted.

[Insert: Correlation Heatmap of Numerical Features]

Fig. 2 — Correlation Heatmap showing strong positive correlation (0.89) between Annual_Spending and Purchase_Frequency, and negative correlation (-0.38) of Discount_Usage_Pct with spending

5.2 Elbow Method Result

The Elbow Method is executed by fitting KMeans for k values from 1 to 10, recording kmeans.inertia_ (WCSS) at each step, and plotting the resulting curve. The curve shows a steep descent from k=1 to k=4, after which the reduction in WCSS becomes much smaller per additional cluster. The visual inflection point at k=4 is clear, indicating four distinct natural groupings in the Annual_Spending vs Purchase_Frequency space. This is not an arbitrary choice — it is the mathematically supported answer to 'how many clusters does this data naturally form?'

[Insert: Elbow Method Graph — WCSS vs k (1 to 10)]

Fig. 3 — Elbow Method showing WCSS reduction across k values. Elbow clearly visible at k=4

5.3 K-Means Cluster Visualization

After fitting KMeans with k=4, cluster assignments are stored in df['Cluster']. Each cluster is visualized as a scatter plot of Annual_Spending vs Purchase_Frequency, with each cluster in a distinct color (#E74C3C, #3498DB, #2ECC71, #F39C12) and centroids marked with black 'X' markers at their inverse-transformed coordinates. The plot visually confirms what the Silhouette Score numerically validates — four clearly separated regions in behavioral space, each representing a distinct purchasing persona.

[Insert: Scatter Plot: K-Means Customer Clusters (k=4) with Centroids]

Fig. 4 — K-Means cluster scatter plot showing four well-separated customer segments in Annual_Spending vs Purchase_Frequency space

5.4 Customer Persona Profiles

After naming the clusters using persona_map, the code generates cluster_profile and persona_summary aggregations using df.groupby('Persona').agg(). These profile tables reveal each persona's average characteristics across all measured features, making the business interpretation concrete:

Persona	Avg Annual Spend	Avg Purchase Freq	Avg Discount %	Avg Returns	Business Interpretation
Budget Buyers	Low (< ₹1,500)	Low (< 5/yr)	High (40%+)	Low (0-1)	Price-sensitive, occasional shoppers.

Persona	Avg Annual Spend	Avg Purchase Freq	Avg Discount %	Avg Returns	Business Interpretation
					Respond to flash sales and discount promotions.
Occasional Shoppers	Moderate (₹1,500–5,000)	Low-moderate (5–12/yr)	Moderate (25–35%)	Low-moderate (1-2)	Infrequent but not absent. Re-engagement campaigns and loyalty point programs work well.
Regular Spenders	Moderate-high (₹5,000–12,000)	Moderate (12–20/yr)	Low (15–25%)	Moderate (2-3)	Consistent buyers. Subscription offers, early access to new products retain this group.
Premium High-Value	High (> ₹12,000)	High (20+/yr)	Very low (< 15%)	Higher (3+)	Top-tier buyers. VIP programs, exclusive launches, and premium service are appropriate.

[Insert: Bar Charts: Average Annual Spending and Purchase Frequency by Persona]

Fig. 5 — Horizontal bar charts comparing average spending and frequency across all four customer personas

5.5 Demographic Analysis — Age Group and Gender

The stacked bar chart of Persona Distribution by Age Group (built using `df.groupby(['Age_Group', 'Persona'])`) reveals how customer personas are distributed across five age bands: 18-25, 26-35, 36-45, 46-55, and 56+. This chart answers questions like 'Are younger customers predominantly Budget Buyers?' and 'Do Premium High-Value Customers skew toward middle-aged groups?' The gender distribution chart similarly shows whether any persona is over-represented by a particular gender, enabling gender-targeted campaign planning.

[Insert: Stacked Bar Chart: Customer Persona Distribution by Age Group]

Fig. 6 — Stacked bar chart showing behavioral persona distribution across age groups (18-25 to 56+)

5.6 Secondary Clustering: Age + Gender (k=3)

The secondary K-Means analysis on `X_ag = df[['Age', 'Gender_Encoded']]` with `k=3` produces three demographic clusters. The scatter plot places Age on the x-axis and Gender_Encoded (0, 1, 2

representing Female, Male, Non-Binary after LabelEncoder) on the y-axis, with centroids marked. The Age-Gender cluster profile table (ag_profile) shows each cluster's average age, dominant gender, and their overlapping behavioral characteristics. The cross-analysis heatmap — `sns.heatmap(pd.crosstab(df['Persona'], df['Age_Gender_Cluster']))` — is the most analytically powerful output of this phase, showing exactly how many customers fall into each combination of behavioral persona and demographic cluster. For example, the heatmap might reveal that 87 customers are simultaneously Premium High-Value Customers AND in the 36-45 female demographic cluster — making them the highest-priority target for a VIP loyalty campaign.

[Insert: Cross-Analysis Heatmap: Persona vs Age-Gender Cluster]

Fig. 7 — Heatmap showing customer count at intersection of each Behavioral Persona and Age-Gender Cluster. Enables hyper-targeted campaign planning.

5.7 Marketing Recommendations by Persona

Persona	Behavioral Signature	Recommended Strategy	Channel
Budget Buyers	Low spending, high discount dependency, infrequent	Flash sales, seasonal discounts, entry-price products	SMS alerts, app push notifications
Occasional Shoppers	Moderate spend, low frequency, moderate discounts	Re-engagement emails, loyalty points, referral bonuses	Weekly newsletters, social media retargeting
Regular Spenders	Consistent purchasing, moderate-high spend	Subscription offers, early product access, bundling	App notifications, personalized email
Premium High-Value	High spending, frequent, minimal discount use	VIP membership, exclusive launches, concierge service	Dedicated account managers, priority support

6. References

Books

- Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow — Aurelien Geron (O'Reilly, 2nd Ed.)
- Python Machine Learning — Sebastian Raschka & Vahid Mirjalili (Packt Publishing, 3rd Ed.)
- Introduction to Machine Learning with Python — Andreas C. Muller & Sarah Guido (O'Reilly Media)
- The Elements of Statistical Learning — Hastie, Tibshirani, Friedman (Springer)

Online Resources

- scikit-learn: KMeans, StandardScaler, silhouette_score — <https://scikit-learn.org/stable/modules/clustering.html>
- Towards Data Science: Customer Segmentation using K-Means Clustering — <https://towardsdatascience.com/>
- GeeksforGeeks: K-Means Clustering in Python — <https://www.geeksforgeeks.org/k-means-clustering-introduction/>
- Kaggle: Retail Customer Segmentation Notebooks — <https://www.kaggle.com/>
- Stack Overflow: Python clustering and visualization — <https://stackoverflow.com>